



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2026

PHYSICS (N530) **TIME: 45 MINUTES**

Candidates answer on the Question Paper.
Additional Materials: Mathematical set, Calculator

Specimen Paper 2026 — For examination practice only. Not an official MES paper.

READ THESE INSTRUCTIONS FIRST

- 1. Write your index number in the space provided above.
- 2. Write in dark blue or black ink. Do not use correction fluid.
- 3. You may use a soft pencil for any diagram, graph or rough working.
- 4. Diagrams are not drawn to scale unless otherwise specified.
- 5. Any rough working should be done in this booklet.
- 6. Answer **ALL** questions.
- 7. This document consists of **6** questions printed on **12** pages, numbered **2** to **13**.
- 8. The total marks for each question is shown in brackets () at the beginning of each question.
- 9. The number of marks for each part question is shown in brackets [].
- 10. The total number of marks for this paper is **50**.

For Examiners' use								
Question No.	1	2	3	4	5	6	Total	Signature
Examiner								
Team Leader								
CE/ACE								

Question 1 (10 marks)**Circle the correct answer. Each item carries one mark.**(a) Which instrument is used to measure **time**?

- A** A ruler
- B** An electronic balance
- C** A stopwatch
- D** A thermometer

(b) What is the S.I. unit of **force**?

- A** kilogram (kg)
- B** joule (J)
- C** newton (N)
- D** watt (W)

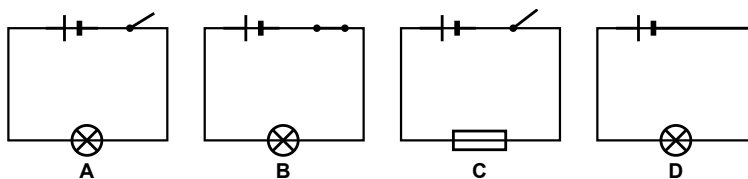
(c) Fig. 1.1 shows four circuit diagrams. Which one shows a **closed** (complete) electric circuit?

Fig. 1.1

(d) An athlete runs a distance of 150 m in 30 s. What is the average **speed** of the athlete?

- A** 0.2 m/s
- B** 5 m/s
- C** 180 m/s
- D** 4500 m/s

(e) Which of the following is a **vector** quantity?

- A** Distance
- B** Mass
- C** Speed
- D** Velocity

(f) A car starts from rest and reaches a speed of 24 m/s in 8 s. What is its **acceleration**?

- A 3 m/s²
- B 16 m/s²
- C 32 m/s²
- D 192 m/s²

(g) Which of the following is a **renewable** source of energy?

- A Coal
- B Oil
- C Wind
- D Natural gas

(h) What is the formula used to calculate the **kinetic energy** of a moving object?

- A mgh
- B $\frac{1}{2}mv^2$
- C ma
- D mv

(i) What is the **unit of electrical resistance**?

- A ampere (A)
- B coulomb (C)
- C ohm (Ω)
- D volt (V)

(j) A current of 2 A flows through a circuit. How much **charge** passes a point in the circuit in 30 s?

- A 15 C
- B 28 C
- C 32 C
- D 60 C

[10]

Question 2 (5 marks)

Fig. 2.1 shows a ray of light striking a plane mirror **MN**. The normal at the point of incidence is shown as a dashed line.

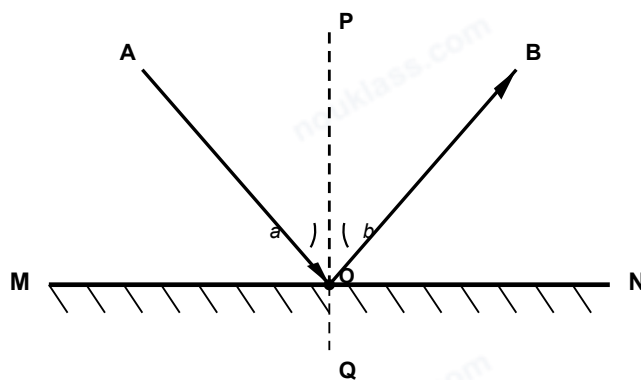


Fig. 2.1

Match each item in **Column A** with its correct label in **Fig. 2.1**.

Column A

- (a) Incident ray
- (b) Angle of incidence
- (c) Reflected ray
- (d) Angle of reflection
- (e) The normal at point O

• •
• •
• •
• •
• •

Column B (from Fig. 2.1)

a
b
AO
OB
PQ

[5]

Question 3 (7 marks)

(a) Define the **time period** of a simple pendulum.

[1]

(b) Fig. 3.1 shows a stopwatch.

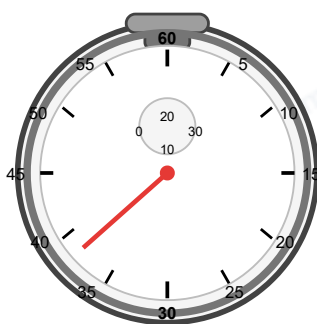


Fig. 3.1

(i) What is the time shown on the stopwatch, in **seconds**?

Time = s [3]

(ii) The time shown in Fig. 3.1 is the time taken for a simple pendulum to complete **20 swings**. Calculate the **time period** of the pendulum. Show all workings.

Time period = s [2]

(iii) Give **one precaution** that should be taken when using the stopwatch to measure the time period.

[1]

Please turn over

Question 4 (10 marks)

(a) Complete the following sentences using the correct word from the list below.

kinetic chemical gravitational potential heat electrical renewable

[Note: there is one extra word in the list]

- (i) A car moving along a road has _____ energy.
- (ii) Food gives us _____ energy.
- (iii) A boulder at the top of a hill has _____ energy.
- (iv) The sun is a _____ source of energy.
- (v) A fan converts _____ energy into _____ energy.

[5]

(b)(i) Which **two** quantities from the list below are **vector** quantities?

Tick (✓) the **two** correct boxes.

☐

Distance

☐

Displacement

☐

Speed

☐

Velocity



Mass

[2]

(b)(ii) A motorcycle accelerates uniformly from 4 m/s to 16 m/s in 6 s.
Calculate the **acceleration** of the motorcycle. Show all your workings.

Acceleration = m/s² [3]

Question 5 (8 marks)

Fig. 5.1 shows a simple series electric circuit containing a cell of e.m.f. 6 V, a switch **S**, an ammeter **A**, and a resistor **R**. Point **P** is marked on the left wire.

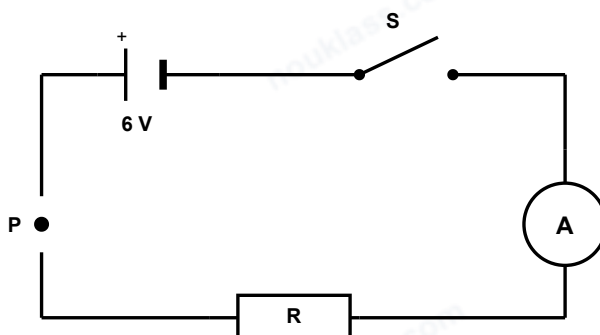


Fig. 5.1

(a) State the **function** of the cell in the circuit.

[1]

(b) In Fig. 5.1, draw an arrow on the wire at point **P** to show the direction of **conventional current**. [1]

(c) The switch **S** is closed and the ammeter reads **0.5 A**.

(i) Calculate the **resistance** of R. Show all your workings.

Resistance = Ω [2]

(ii) The switch is closed for **4 minutes**. Calculate the total **charge** that flows through the circuit. Show all workings.

Charge = C [2]

(iii) How much **electrical energy** is converted by the resistor R in 4 minutes? Show all workings.

Energy = J [2]

Question 6 (10 marks)

Fig. 6.1 shows the speed-time graph of a bus during part of its journey.

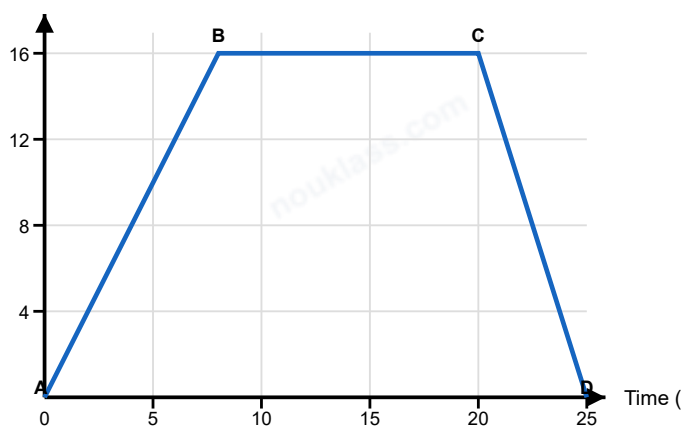


Fig. 6.1

(a)

(i) For how long does the bus move at **constant speed**?

Time = s [1]

(ii) What is the **maximum speed** reached by the bus?

Maximum speed = m/s [1]

(b) Fully describe the motion of the bus between

(i) A and B.

.....

(ii) C and D.

.....

[1]

[1]

(c) Calculate the **acceleration** of the bus between A and B. Show all your workings.

Acceleration = m/s² [2]

(d) Calculate the **total distance** travelled by the bus during the whole journey shown in Fig. 6.1.
Show all your workings.

Total distance = m [2]

(e) The mass of the bus is **5000 kg**. Calculate the **force** needed to produce the acceleration between A and B. Show all your workings. [Take $F = ma$]

Force = N [2]

Specimen Mark Scheme — Paper 1 of 2

Question 1 — Multiple Choice (10 marks, 1 each)

- (a) C — stopwatch
- (b) C — newton (N)
- (c) B — closed circuit with switch closed (all components connected with switch bridging the gap)
- (d) B — 5 m/s $v = d/t = 150/30 = 5 \text{ m/s}$
- (e) D — Velocity (has both magnitude and direction)
- (f) A — 3 m/s² $a = (v-u)/t = (24-0)/8 = 3 \text{ m/s}^2$
- (g) C — Wind (renewable: replenished naturally)
- (h) B — $\frac{1}{2}mv^2$
- (i) C — ohm (Ω)
- (j) D — 60 C $Q = It = 2 \times 30 = 60 \text{ C}$

Question 2 — Reflection Matching (5 marks)

- (a) Incident ray → **AO** [1]
- (b) Angle of incidence → **a** [1]
- (c) Reflected ray → **OB** [1]
- (d) Angle of reflection → **b** [1]
- (e) The normal at point O → **PQ** [1]

Question 3 — Pendulum and Stopwatch (7 marks)

- (a) The time period is the time taken for the pendulum to complete **one complete oscillation** (swing to and fro). [1]
- (b)(i) The stopwatch reads **38 s**. Award [3] marks: [1] for reading to the nearest 5 s (35–40), [1] for reading to the nearest second, [1] for correct value 38 s.
- (b)(ii) Formula: $T = \text{total time} / \text{number of swings} = 38 / 20 = 1.9 \text{ s}$ [2: formula/substitution [1], answer with unit [1]]
- (b)(iii) Accept any one of: Start and stop the stopwatch exactly when the pendulum passes through the lowest point / count oscillations carefully / repeat and average / reduce parallax error. [1]

Question 4 — Energy and Motion (10 marks)

- (a)(i) kinetic [1]
- (a)(ii) chemical [1]
- (a)(iii) gravitational potential [1]
- (a)(iv) renewable [1]
- (a)(v) electrical ... kinetic [1]
- (b)(i) Tick: **Displacement** and **Velocity** [2]
- (b)(ii) $a = (v - u) / t = (16 - 4) / 6 = 12 / 6 = 2 \text{ m/s}^2$
Formula stated or implied [1], correct substitution [1], answer = 2 m/s² with unit [1]

Question 5 — Electricity (8 marks)

- (a) The cell/battery provides electrical energy / drives current around the circuit. [1]
- (b) Arrow pointing **downward** on the left wire at P (conventional current flows from + terminal, clockwise in this circuit: down the left side). [1]
- (c)(i) $V = IR \rightarrow R = V/I = 6 / 0.5 = 12 \Omega$ [formula [1], answer with unit [1]]
- (c)(ii) Convert 4 min to seconds: $4 \times 60 = 240 \text{ s}$
 $Q = I \times t = 0.5 \times 240 = 120 \text{ C}$ [conversion [1], answer with unit [1]]
- (c)(iii) $E = VIt = 6 \times 0.5 \times 240 = 720 \text{ J}$ (or $E = QV = 120 \times 6 = 720 \text{ J}$) [formula/substitution [1], answer with unit [1]]

Question 6 — Speed-Time Graph (10 marks)

- (a)(i) Bus at constant speed between B and C: $20 - 8 = 12 \text{ s}$ [1]
(a)(ii) Maximum speed = **16 m/s** [1]
(b)(i) Between A and B: The bus **accelerates uniformly** from rest (0 m/s) to 16 m/s. [1]
(b)(ii) Between C and D: The bus **decelerates uniformly** from 16 m/s to rest. [1]
(c) $a = (v - u) / t = (16 - 0) / 8 = 2 \text{ m/s}^2$ [formula/substitution [1], answer with unit [1]]
(d) Total distance = area under graph:
 $= \frac{1}{2} \times 8 \times 16 + 12 \times 16 + \frac{1}{2} \times 5 \times 16$
 $= 64 + 192 + 40 = 296 \text{ m}$ [correct method [1], answer with unit [1]]
(e) $F = ma = 5000 \times 2 = 10\,000 \text{ N}$ [formula/substitution [1], answer with unit [1]]

** This specimen mark scheme is for practice purposes only. The real NCE 2026 paper and mark scheme are set exclusively by the Mauritius Examinations Syndicate (MES). This paper is produced by Noukclass.com.*

End of Paper 1